Model Architecture

1 Interpretation Network

To interpret the prediction network, we train a small auxiliary network that operates at an intermediate convolutional layer. Let $\mathbf{z} \in \mathbb{R}^{\ell \times c}$ denote the code (feature map) at the chosen layer for a given input sequence, and let $f(\mathbf{z}) \in \mathbb{R}$ denote the scalar output of the prediction network from that layer onward.

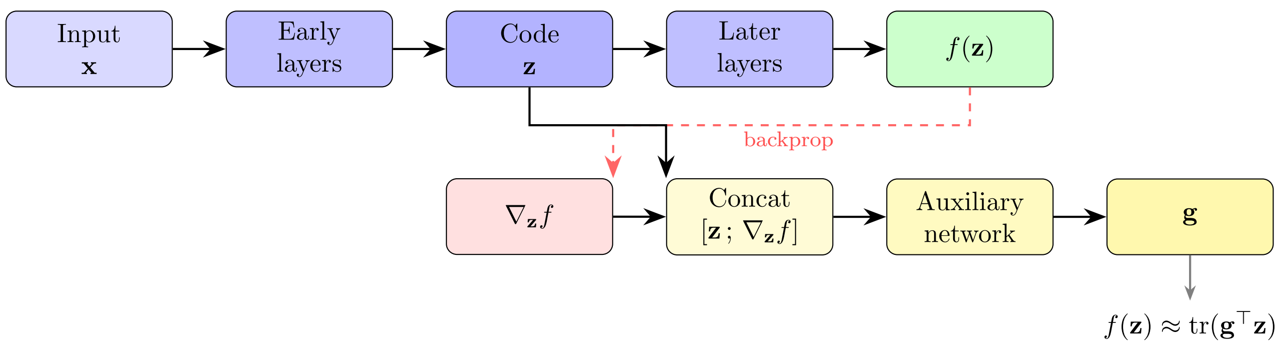

Figure 1: Interpretation network. The auxiliary network receives the code $\mathbf{z}$ and its gradient $\nabla_{\mathbf{z}} f$, and outputs a pseudo-gradient $\mathbf{g}$ that linearly approximates the prediction.

The auxiliary network takes as input both $\mathbf{z}$ and the gradient $\nabla_{\mathbf{z}} f$ (concatenated along the channel axis) and produces a pseudo-gradient $\mathbf{g} \in \mathbb{R}^{\ell \times c}$ of the same shape as $\mathbf{z}$. It is trained so that the prediction network's output is well-approximated by the linear surrogate

$$f(\mathbf{z}) \;\approx\; \mathrm{tr}\big(\mathbf{g}^\top \mathbf{z}\big) \;=\; \sum_{i,j} g_{ij}\, z_{ij}\,. \tag{1}$$

Because $\mathbf{g}$ has the same spatial and channel dimensions as the code, each entry g_{ij} quantifies the local importance of the corresponding feature at position i and channel j. This provides a per-position, per-filter attribution that can be read off directly from the learned convolutional features, enabling downstream motif analysis without relying on post-hoc gradient saliency alone.

2 Prediction Network

The prediction network is a deep neural network that maps one-hot encoded biological sequences to scalar (or multi-output) predictions. It combines convolutional feature extraction with channel-wise attention, drawing on design principles from EfficientNet [2, 3], and operates in two stages.

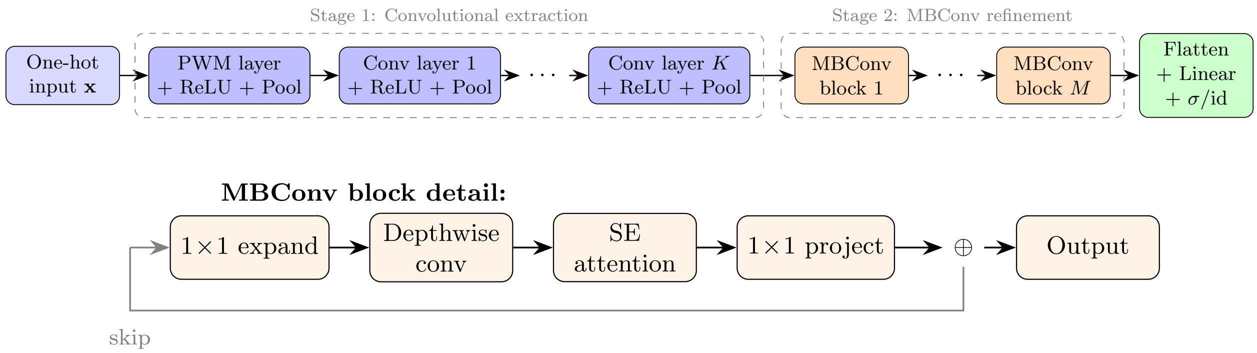

Figure 2: Prediction network architecture. Top: overall pipeline. Bottom: detail of each MBConv block with residual skip connection.

Stage 1: Convolutional feature extraction. The first layer consists of learnable position weight matrices (PWMs) applied directly to the one-hot input, followed by ReLU activation and pooling. Subsequent convolutional layers operate on the resulting feature maps with additional pooling, progressively reducing spatial resolution while increasing representational capacity. All convolutional layers in this stage use standard (non-depthwise) convolutions, so the network interacts with the raw sequence signal immediately at its input.

Stage 2: MBConv refinement. After the feature-extraction stage, the representations pass through a series of Mobile Inverted Bottleneck Convolution (MBConv) blocks [2]. Each block interleaves local convolution with squeeze-and-excitation (SE) channel attention: (i) 1×1 channel expansion, (ii) depthwise convolution, (iii) SE attention that recalibrates channel responses globally, and (iv) 1×1 projection back to the original channel count, with a residual skip connection. This hybrid convolution–attention design refines the learned representations without substantially increasing the parameter count. Similar architectures combining convolutional layers with channel attention have been adopted for sequence-to-function prediction tasks and demonstrated state-of-the-art performance [1].

Output layer. The refined feature maps are flattened into an embedding vector and projected to the output dimension via a linear layer. Depending on the task, the final nonlinearity is either the identity (regression) or the element-wise sigmoid (classification / probability outputs).

References

[1] Dmitry Penzar, Daria Nogina, Elizaveta Noskova, Arsenii Zinkevich, Georgy Meshcheryakov, Andrey Lando, Abdul Muntakim Rafi, Carl De Boer, and Ivan V Kulakovskiy. Legnet: a best-in-class deep learning model for short dna regulatory regions. *Bioinformatics*, 39(8):btad457, 2023.

[2] Mingxing Tan and Quoc Le. Efficientnet: Rethinking model scaling for convolutional neural networks. In *International conference on machine learning*, pages 6105–6114. PMLR, 2019.

[3] Mingxing Tan and Quoc Le. Efficientnetv2: Smaller models and faster training. In *International conference on machine learning*, pages 10096–10106. PMLR, 2021.